

ANDREW TAN

Mechanical & Space Engineering Student | +61 426 534 677 | andrew_tan1@icloud.com | Sydney, NSW

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Engineering Honours (Mechanical with Space)

Sydney, NSW

Feb 2024 – Nov 2027

- Dalyell Scholar, 83.25% overall WAM (Distinction), SUGS Subcommittee and Undergraduate Engineering Mentor
- Relevant Coursework: Fluid Mechanics, Control Systems and Numerical Methods, Engineering Analysis, Systems Design, Thermodynamics, Mechanics and Dynamics, Computing and Multi-disciplinary Engineering, Vibration and Heat Transfer

THE SCOTS COLLEGE

High School Certificate

Sydney, NSW

2009 — 2023

- ATAR: 98.95 | HSC All-rounder | Maths Extension 1 & 2, Physics, Chemistry, Advanced English
- Activities: AAGPS 1st XI football, Ashburner society, F1 in Schools, Pipes and Drums, Accelerated Mathematics Extension 2

WORK EXPERIENCE

USYD ROCKETRY TEAM

Senior Flight Systems Engineer

Sydney, NSW

Aug 2025 — Present

- Presented flight systems design at a Critical Design Review to external stakeholders, consolidating structural analysis and interface documentation across propulsion, aerostructures and recovery subsystems, securing supplier and machining support.
- Modelled vehicle aerodynamic stability and drag in OpenRocket to inform fin geometry and apogee prediction, iterating configurations against flight performance requirements.
- Designed aft truss, adapter plates, and bulkheads in Onshape, driving material selection through FEA; authored ANSYS stress-analysis documentation and ran mesh convergence studies to validate results.
- Executed destructive testing of aluminium members on static and dynamic machines, identifying failure modes (yield, buckling, shear-out) and comparing measured failure loads against FEA predictions.

ALUMICORP

CNC Production Intern

Sydney, NSW

July 2026 — Present

- Set up and operated 3-axis CNC mills and a laser cutter for aluminium production runs, applying tool and work offsets and running inspection with calipers and micrometers, reporting measured dimensions against drawing tolerances.
- Interpreted and completed GD&T engineering drawings, editing inconsistencies for encoding, tuning feeds, speeds, and work holding, manoeuvring aluminium sheets, and authoring cutting performance reporting.

ATLAS TUTORING COLLEGE

Academic Tutor (K to Year 12 One on One & group tuition)

Sydney, NSW

Feb 2024 — Present

- Delivered specialised tuition (Years 10–12) in Physics, Chemistry and Maths, adapting lesson structure to improve results.

THE SCOTS COLLEGE

Football Coach

Sydney, NSW

Jan 2024 — Present

- Assistant coach for 1st and 2nd XI squads, leading training for Years 7–12 and fostering player development and team culture.

UNIVERSITY PROJECTS

MAGLEV LEVITATION CONTROL

Designed a state-feedback pole-placement controller with Luenberger observer for an unstable EMS maglev system, validated in MATLAB Simulink across reference tracking, mass uncertainty, and disturbance rejection scenarios.

May 2026

SPACE MISSION MODELLING

Built Python software implementing orbital mechanics and coordinate-transform routines across three mission cases, structuring the code for reuse, and validating numerical outputs against analytical solutions.

Aug 2025

PNEUMATIC ENGINE

Worked to produce a dual-acting pneumatic engine combining a flywheel and sliding-valve system, performing material selection, stress, and tolerance analysis, and producing SolidWorks CAD models and engineering drawings. Gained hands-on lab experience through operating the lathe, mill, and drill press to machine custom components.

Sep 2024

SKILLS AND ACTIVITIES

Technical Skills: Python, MATLAB/Simulink, ANSYS, Solidworks, Onshape, OpenRocket, Excel

Additional: New Zealand, Dutch and Australian citizenship, Bronze and Silver DOE | Oporto, Rebel Sport (customer-facing)

Interests: ESFA Division 3 Football, Chess (1650 rating), Dutch (limited proficiency), and Surfing.